

AI-Powered CRM Platform - System Analysis Document

CNC111 - Web Programming Project

Team Members

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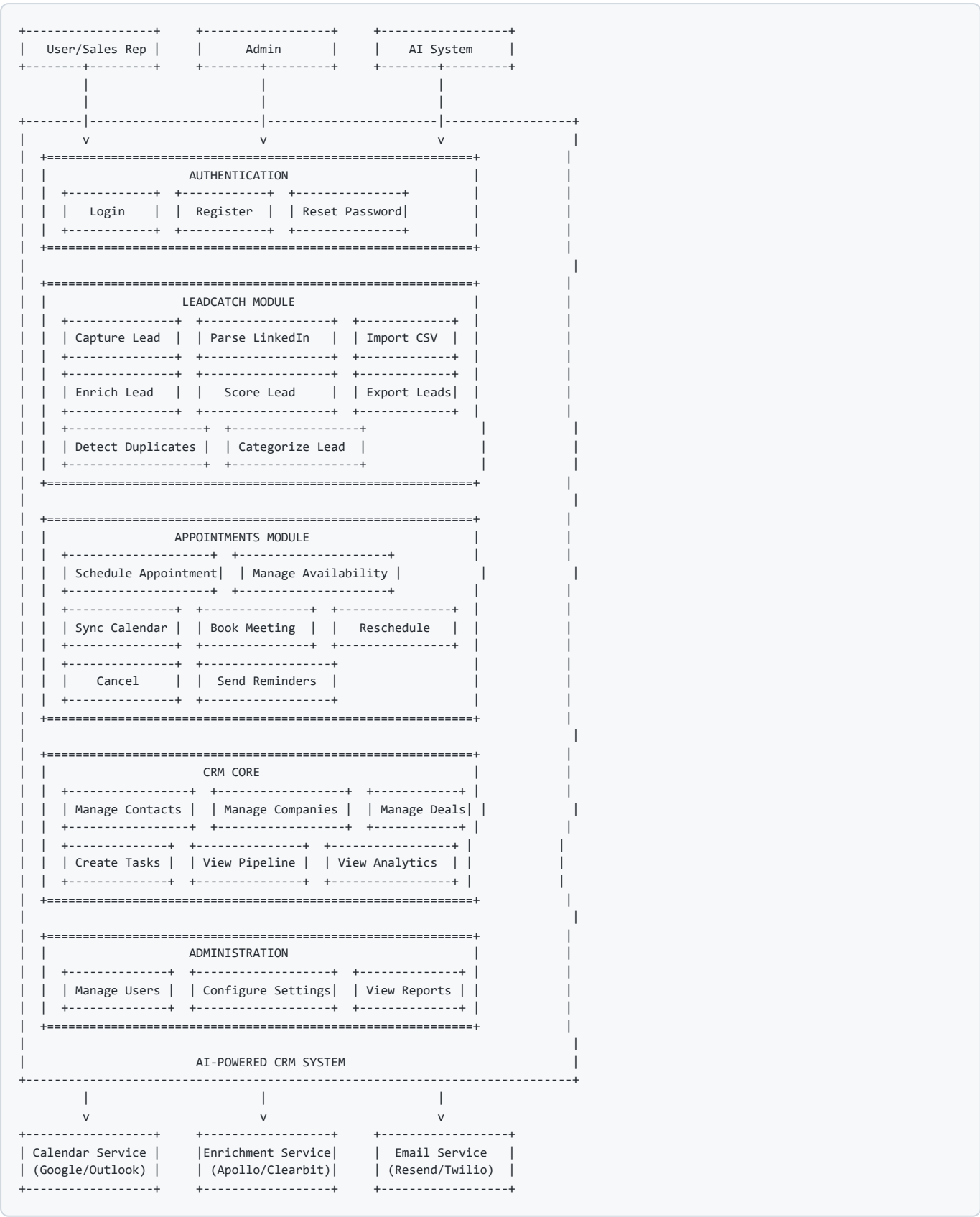
System Overview

The **AI-Powered CRM Platform** is a modern Customer Relationship Management system that leverages artificial intelligence to enhance lead management, appointment scheduling, and customer interactions. The system is built using Next.js 14+, Supabase, and integrates with OpenAI GPT-4o for intelligent automation.

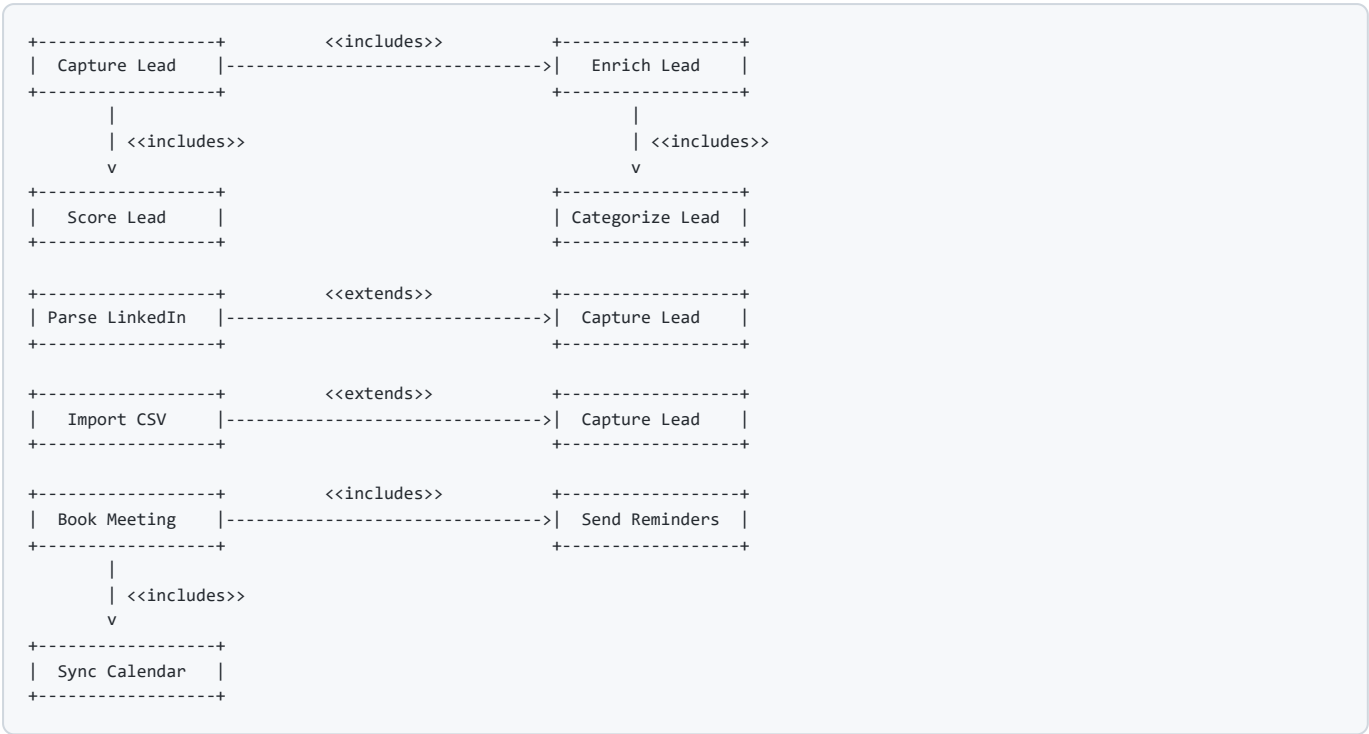
Key Modules:

- **LeadCatch Module** - AI-powered lead capture, parsing, and enrichment
- **Appointments Module** - Smart scheduling with calendar integrations
- **CRM Core** - Contact management, sales pipeline, and analytics
- **AI Module** - Shared AI services for all modules

Use Case Diagram



Use Case Relationships



Narrative Use Cases

Use Case 1: Capture and Enrich Lead

Field	Description
Use Case ID	UC-001
Use Case Name	Capture and Enrich Lead
Actor(s)	User (Sales Rep), AI System, Enrichment Service
Description	User captures a lead from various sources, and the system automatically enriches it with additional data using AI
Preconditions	User is logged into the system
Postconditions	Lead is stored in database with enriched data and AI-generated score

Main Flow:

Step	Action
1	User accesses the LeadCatch module
2	User selects input method (LinkedIn URL, Website, Manual, or CSV)
3	User provides lead information
4	System validates input data
5	AI parses and extracts lead information
6	System initiates waterfall enrichment (Apollo → Clearbit)
7	AI scores the lead (1-100)
8	AI categorizes and tags the lead
9	System checks for duplicates
10	System saves the lead to database
11	System displays enriched lead profile to user

Alternative Flows:

Step	Condition	Action
4a	Invalid input	System displays error, returns to step 3
6a	Service unavailable	System saves partial data, flags for retry
9a	Duplicate found	System prompts user to merge or create new

Use Case 2: Schedule Appointment

Field	Description
Use Case ID	UC-002
Use Case Name	Schedule Appointment
Actor(s)	User, External Attendee, Calendar Service, AI System
Description	User or external attendee schedules a meeting through the system
Preconditions	User has configured availability and event types
Postconditions	Appointment is created and synced to all calendars

Main Flow:

Step	Action
1	Attendee accesses public booking page
2	Attendee selects event type
3	System retrieves user's availability from connected calendars
4	System calculates available time slots
5	Attendee selects preferred date and time
6	Attendee enters contact information
7	AI suggests optimal meeting time (optional)
8	System validates availability (no conflicts)
9	System creates appointment in database
10	System syncs to Google Calendar/Outlook
11	System sends confirmation email to both parties
12	System schedules reminder notifications

Alternative Flows:

Step	Condition	Action
5a	No available slots	System suggests alternative dates
8a	Conflict detected	Returns to step 5 with updated availability
10a	Calendar sync fails	System queues for retry, notifies user

Use Case 3: Manage Sales Pipeline

Field	Description
Use Case ID	UC-003
Use Case Name	Manage Sales Pipeline
Actor(s)	User (Sales Rep), Admin
Description	User manages deals through different stages of sales pipeline
Preconditions	User is logged in, deals exist in the system
Postconditions	Deal status is updated, activities are logged

Main Flow:

Step	Action
1	User accesses Pipeline view (Kanban board)
2	System displays deals organized by stage
3	User drags deal card to new stage
4	System updates deal status
5	System logs activity in timeline
6	System triggers automated actions (notifications, tasks)
7	System updates analytics

Use Case 4: Import Leads from CSV

Field	Description
Use Case ID	UC-004
Use Case Name	Import Leads from CSV
Actor(s)	User, AI System
Description	User bulk imports leads from a CSV file
Preconditions	User has a CSV file with lead data
Postconditions	All valid leads are imported and enriched

Main Flow:

Step	Action
1	User accesses Import function in LeadCatch
2	User uploads CSV file
3	System validates file format
4	System displays field mapping interface
5	User maps CSV columns to lead fields
6	System validates data in each row
7	System detects duplicates
8	User confirms import (with duplicate handling options)
9	System imports leads in batches
10	System queues leads for AI enrichment
11	System displays import summary report

Use Case 5: Generate AI-Powered Email

Field	Description
Use Case ID	UC-005
Use Case Name	Generate AI-Powered Email
Actor(s)	User, AI System
Description	User generates personalized email content using AI
Preconditions	User has selected a lead or contact
Postconditions	Email is generated and ready to send

Main Flow:

Step	Action
1	User selects a lead/contact
2	User clicks "Generate Email"
3	User selects email type (intro, follow-up, proposal, etc.)
4	System retrieves lead context (company, role, interactions)
5	AI generates personalized email draft
6	System displays draft to user
7	User reviews and edits
8	User sends or saves draft

Entity Relationship Diagram (ERD)

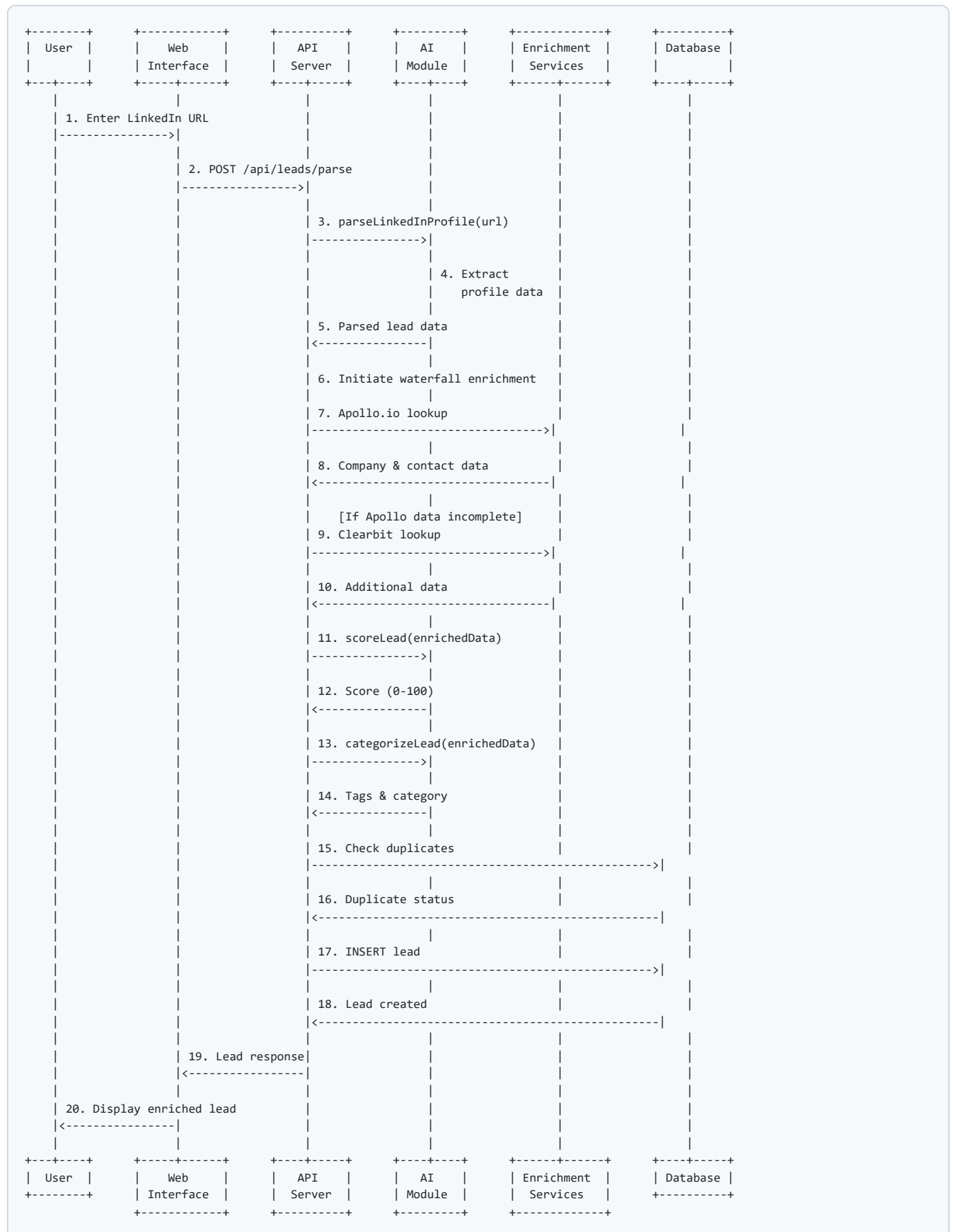
		EVENT_TYPES		
		+-----+		
		PK: id (uuid)		
		FK: user_id		
		name		
		slug (unique)		
		description		
		duration_minutes		
		color		
		is_active		
		availability_rules		
		buffer_times		
		created_at		
		updated_at		
		+-----+		
+-----+				
SUPPORTING ENTITIES				
+-----+				
+-----+		+-----+		+-----+
TASKS		ACTIVITIES		NOTES
+-----+		+-----+		+-----+
PK: id (uuid)		PK: id (uuid)		PK: id (uuid)
FK: user_id		FK: user_id		FK: user_id
FK: contact_id		FK: contact_id		FK: contact_id
FK: deal_id		FK: lead_id		FK: lead_id
title		FK: deal_id		FK: deal_id
description		type		content
priority		title		is_pinned
status		description		created_at
due_date		metadata (json)		updated_at
completed_at		created_at		+-----+
created_at		+-----+		
updated_at				
+-----+				
+-----+		+-----+		+-----+
AVAILABILITY		CALENDAR_CONNECTIONS		AI_LOGS
+-----+		+-----+		+-----+
PK: id (uuid)		PK: id (uuid)		PK: id (uuid)
FK: user_id		FK: user_id		FK: user_id
day_of_week		provider		action_type
start_time		access_token		model_used
end_time		refresh_token		tokens_used
is_available		calendar_id		input_data (json)
created_at		token_expires_at		output_data(json)
+-----+		is_active		response_time_ms
		created_at		created_at
		updated_at		+-----+
		+-----+		
+-----+		+-----+		+-----+
TAGS		LEAD_TAGS		CONTACT_TAGS
+-----+		+-----+		+-----+
PK: id (uuid)	<---	FK: tag_id		FK: contact_id
FK: user_id		FK: lead_id		FK: tag_id
name		+-----+		+-----+
color				
entity_type				
created_at				
+-----+				

ERD Relationships Summary

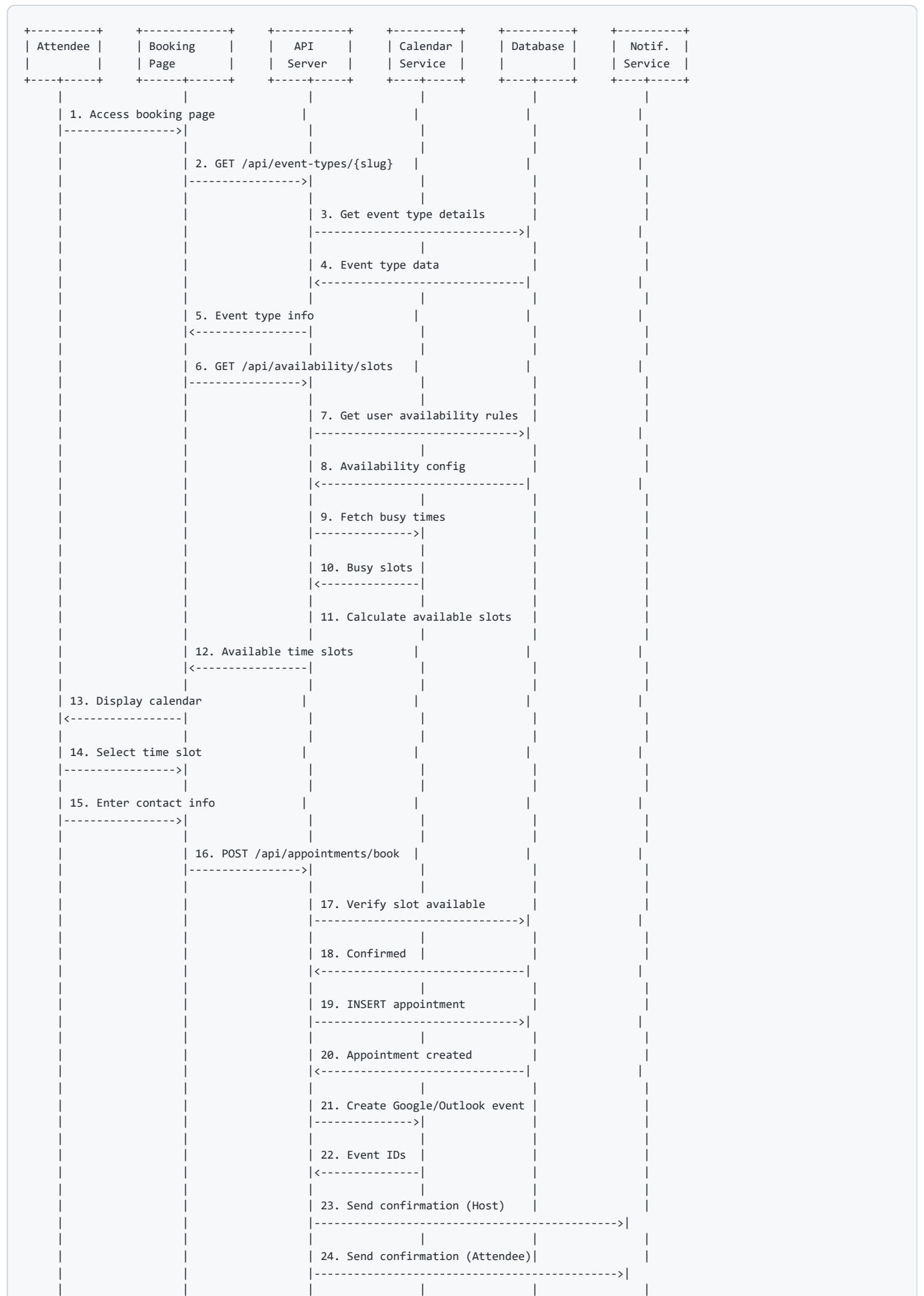
Relationship	Type	Description
Users → Leads	1 : N	User owns many leads
Users → Contacts	1 : N	User owns many contacts
Users → Companies	1 : N	User owns many companies
Users → Deals	1 : N	User owns many deals
Users → Appointments	1 : N	User hosts many appointments
Users → Event_Types	1 : N	User creates many event types
Users → Tasks	1 : N	User creates many tasks
Companies → Contacts	1 : N	Company employs many contacts
Companies → Deals	1 : N	Company associated with many deals
Contacts → Deals	1 : N	Contact involved in many deals
Contacts → Appointments	1 : N	Contact attends many appointments
Event_Types → Appointments	1 : N	Event type defines many appointments
Leads → Lead_Tags	1 : N	Lead has many tags
Contacts → Contact_Tags	1 : N	Contact has many tags
Tags → Lead_Tags	1 : N	Tag applied to many leads
Tags → Contact_Tags	1 : N	Tag applied to many contacts

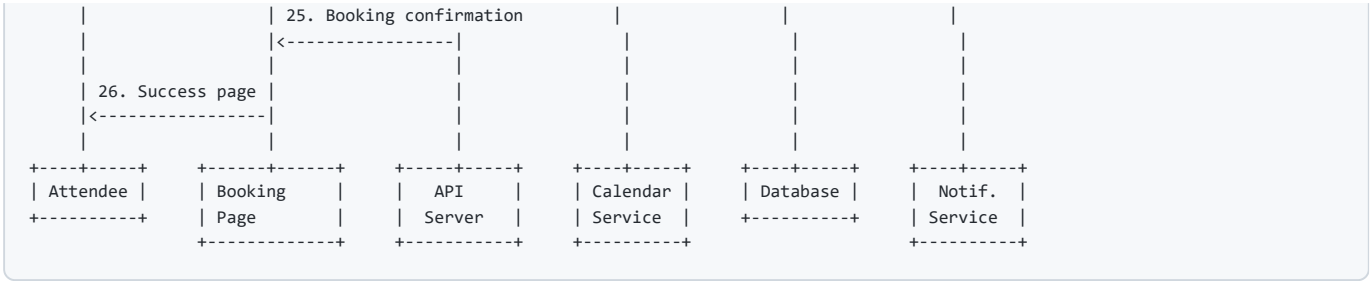
Sequence Diagrams

Sequence Diagram 1: Lead Capture and Enrichment

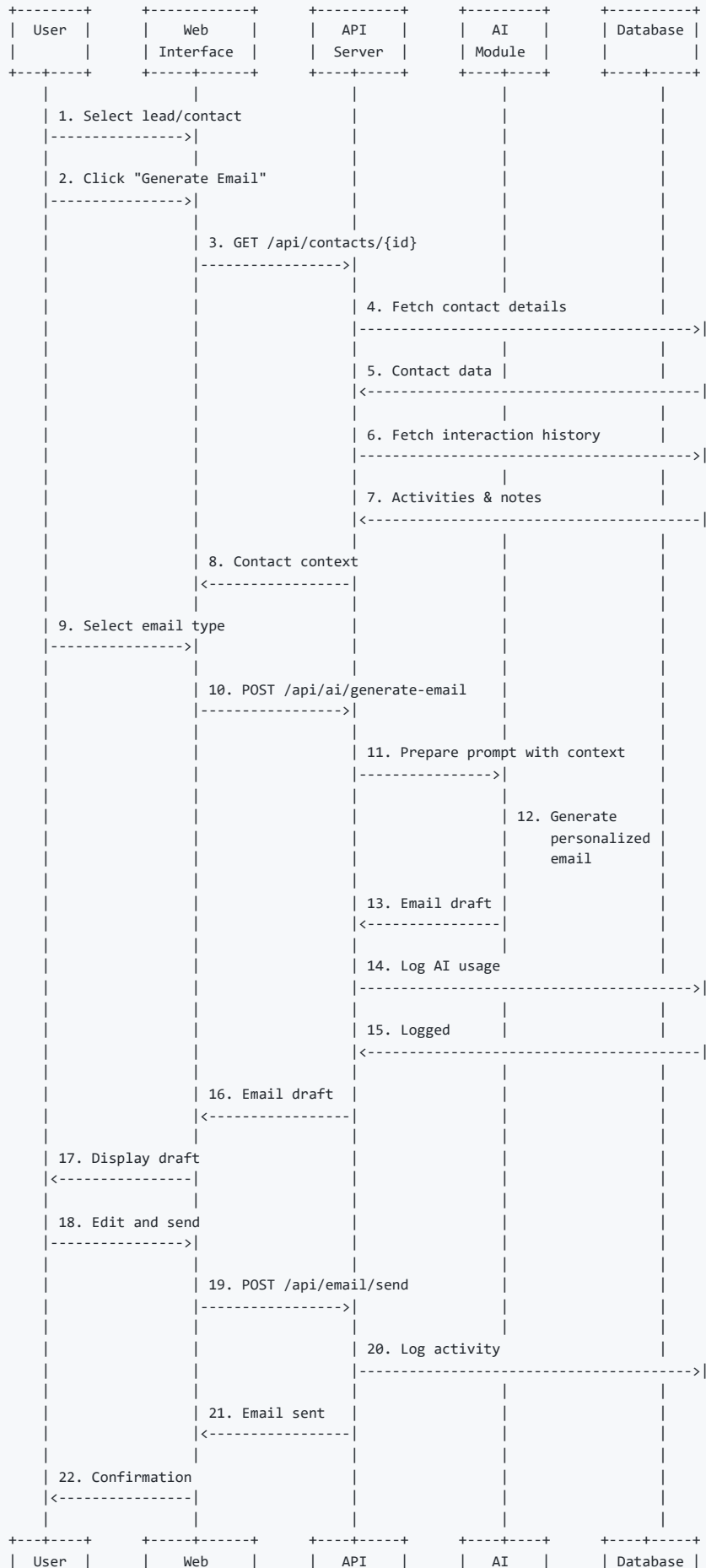


Sequence Diagram 2: Appointment Booking



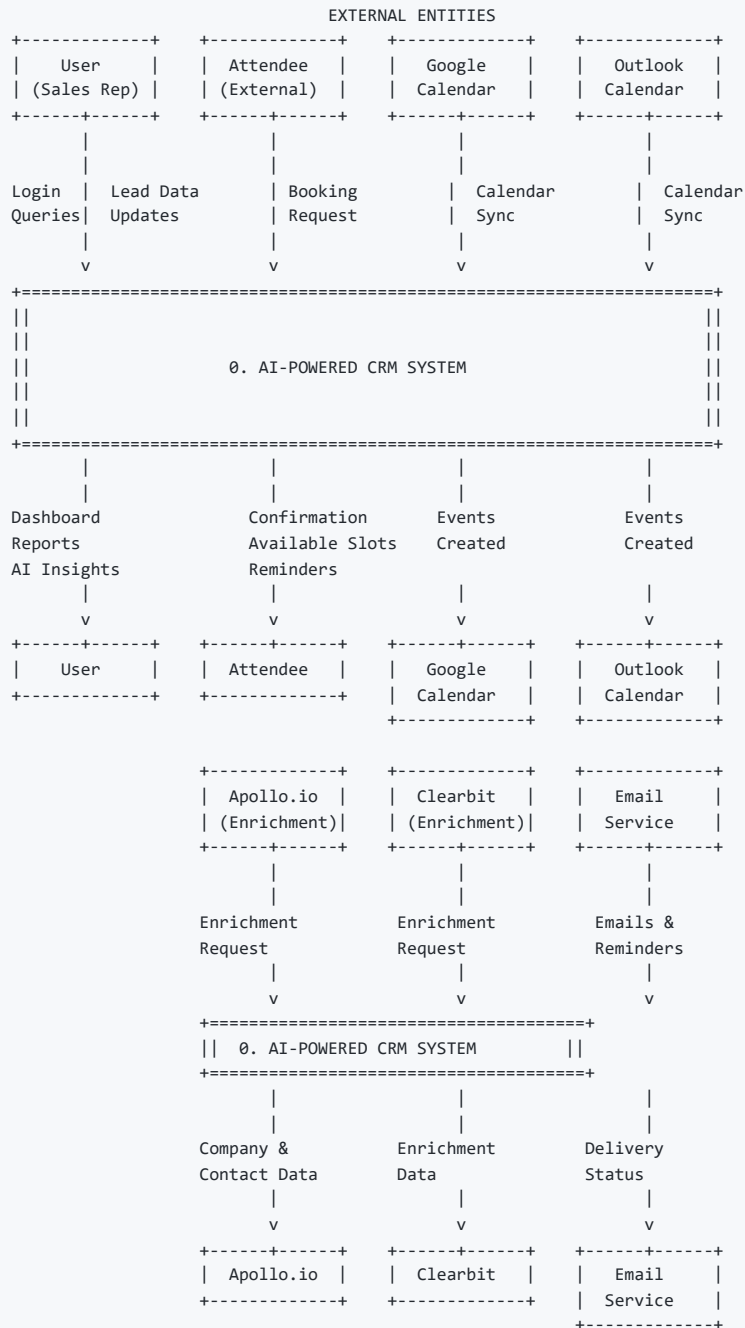


Sequence Diagram 3: AI Email Generation



Data Flow Diagrams (DFD)

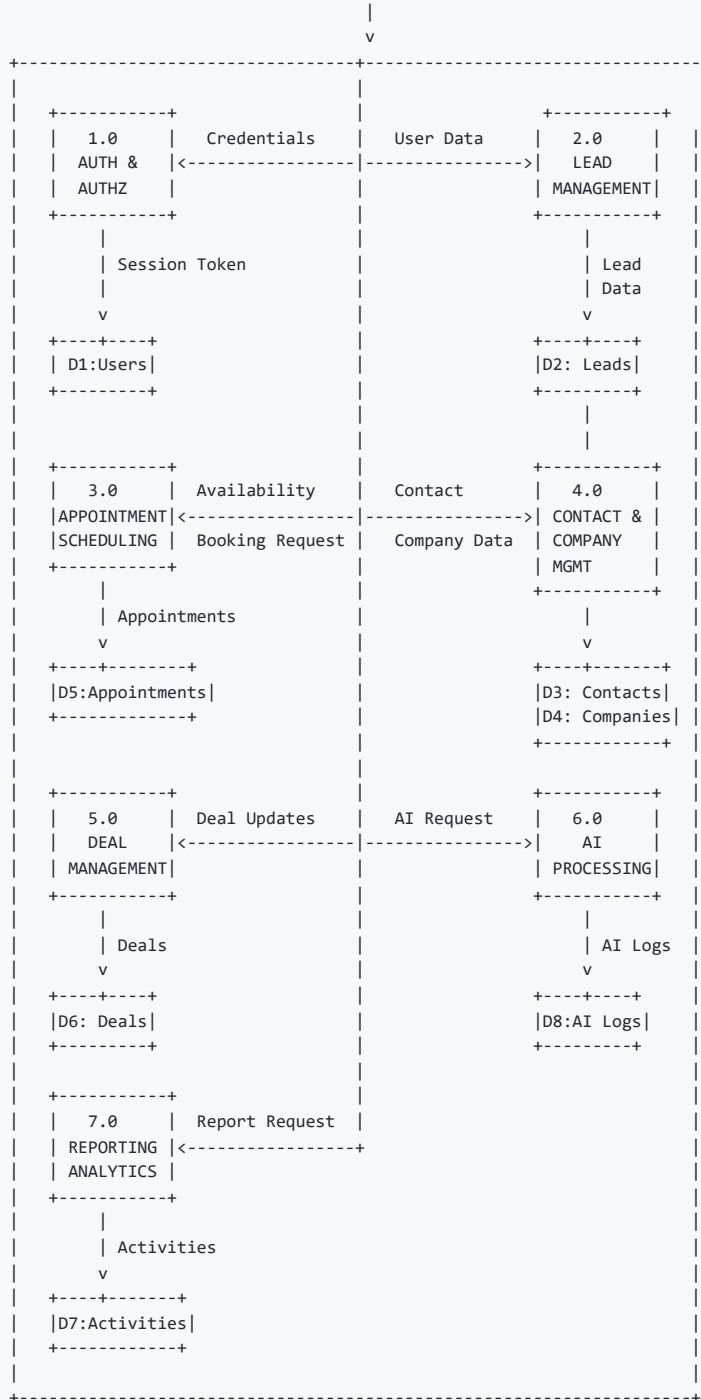
Level 0: Context Diagram



Level 1: Main Processes

```
+=====+
|                                     |
|                               DFD LEVEL 1 - MAIN PROCESSES                               |
|                                     |
+=====+
```

EXTERNAL ENTITIES:
[User] [Attendee] [Google Calendar] [Outlook] [Enrichment APIs] [Email]

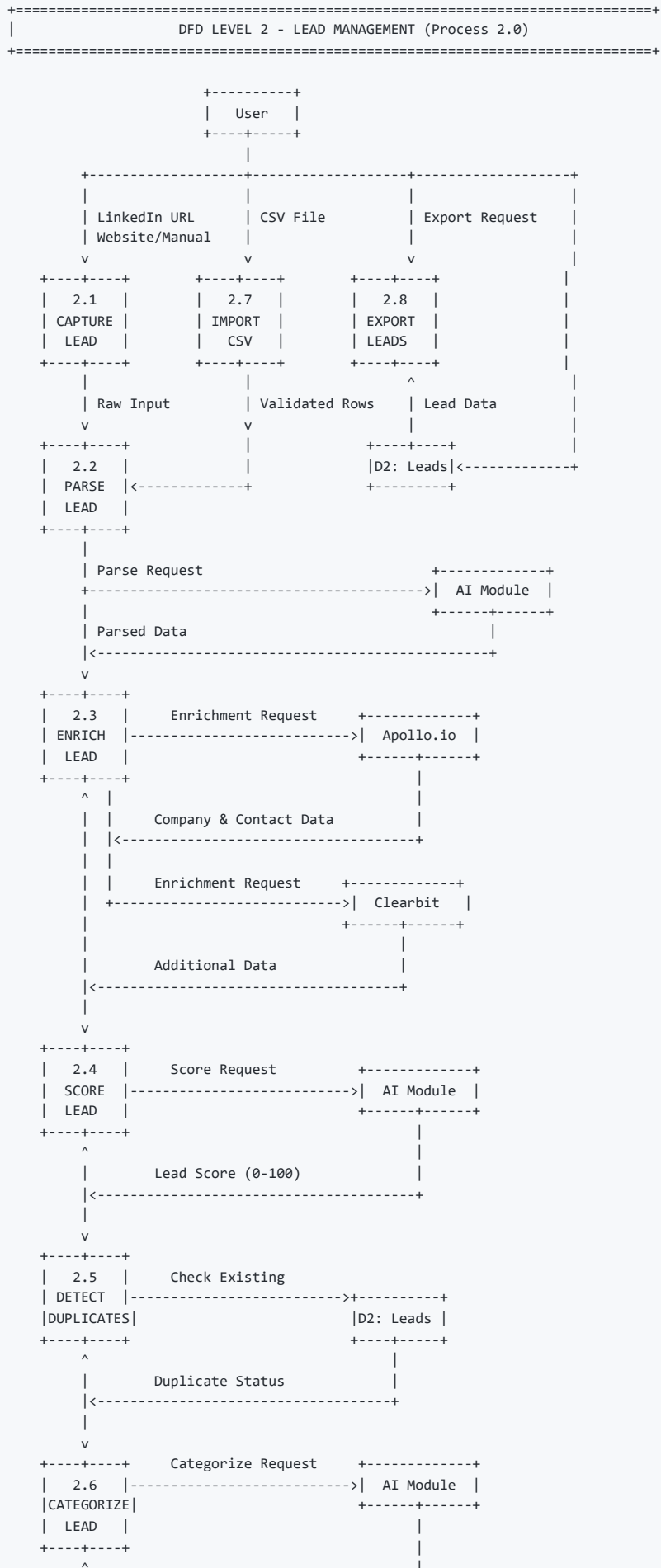


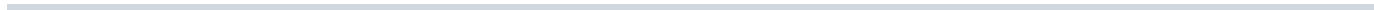
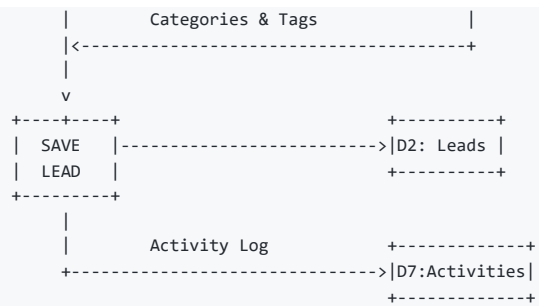
DATA STORES:

```
+-----+ +-----+ +-----+ +-----+
| D1:Users | | D2:Leads | | D3:Contacts | | D4:Companies |
+-----+ +-----+ +-----+ +-----+
```

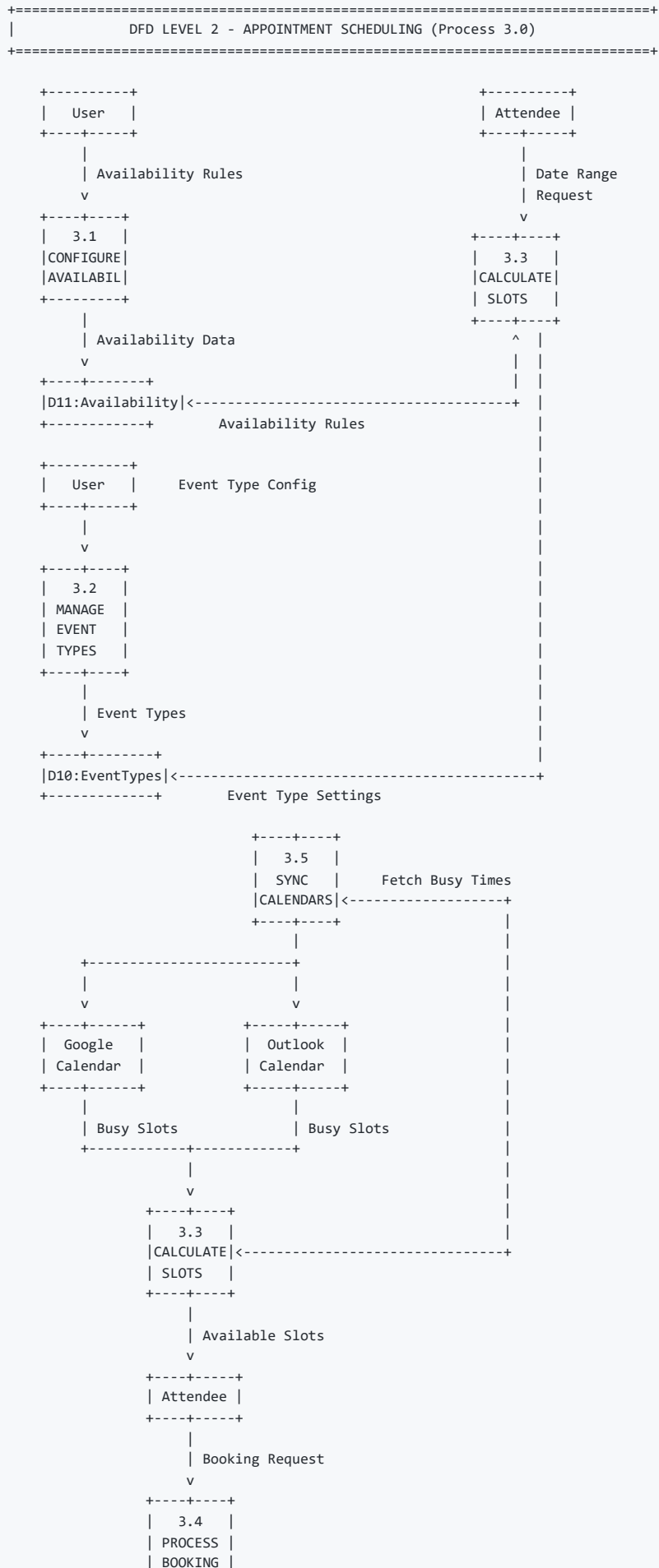
```
+-----+ +-----+ +-----+ +-----+
| D5:Appointments | | D6:Deals | | D7:Activities | | D8:AI Logs |
+-----+ +-----+ +-----+ +-----+
```

Level 2: Lead Management Process (2.0)

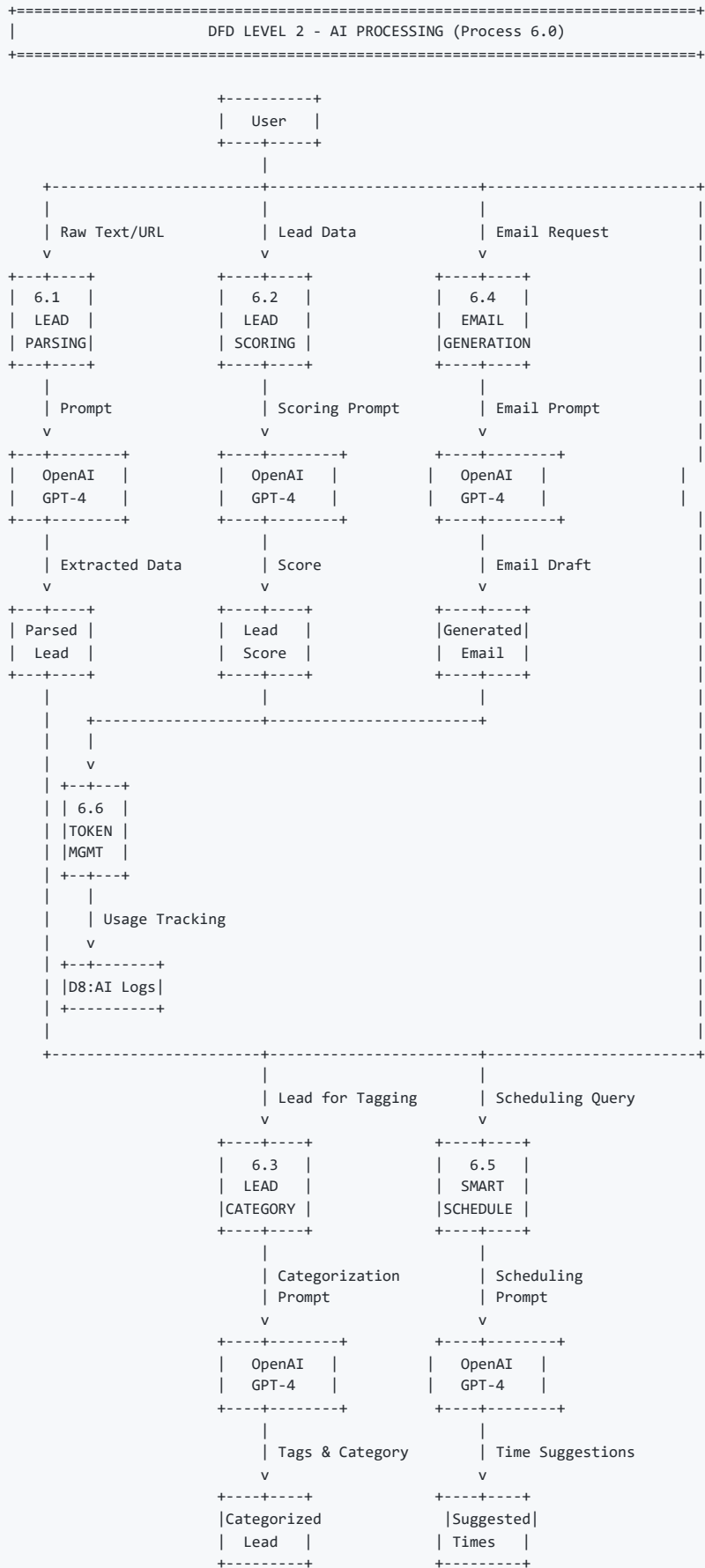




Level 2: Appointment Scheduling Process (3.0)



Level 2: AI Processing (6.0)



Technology Stack Summary

Layer	Technology
Frontend	Next.js 14+, React 18+, TypeScript, Tailwind, Shadcn
Backend	Next.js API Routes, Drizzle ORM
Database	PostgreSQL (Supabase)
Authentication	Supabase Auth
AI	OpenAI GPT-4o
Calendar	Google Calendar API, Microsoft Outlook API
Enrichment	Apollo.io , Clearbit
Notifications	Resend/SendGrid (Email), Twilio (SMS)
Deployment	Vercel

References

- Next.js Documentation: <https://nextjs.org/docs>
- Supabase Documentation: <https://supabase.com/docs>
- OpenAI API Reference: <https://platform.openai.com/docs>
- Drizzle ORM Documentation: <https://orm.drizzle.team/docs>
- Google Calendar API: <https://developers.google.com/calendar>
- Microsoft Graph API: <https://docs.microsoft.com/en-us/graph>

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